

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	B	D	D	A	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C	A,C,D	A,B	A,B,D	A,C,D	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.28	3.00	1.90	34.58	1.98	19.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	C	A	B	A	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C,D	A,B,D	A,B,C,D	A,C,D	A,B,C	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	11.00	5.00	6.00	990.00	7.00	2.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	C	B	A	A	A
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C	A,B,C,D	A,B	A,C	A,B,C,D	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.00	5.00	7.75	20.00	2.00	45.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

 1. **Ans (A)**

$$(20 + 10) \frac{1 \times 10^{-3}}{2} = s \times \left(10 - \frac{1 \times 10^{-3}}{2} \right)$$

$$\Rightarrow s \simeq 1.5 \text{ m}\Omega$$

 2. **Ans (B)**

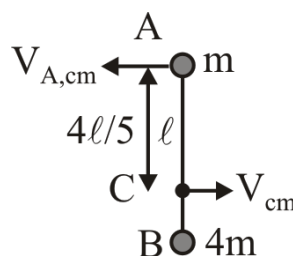
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_1 = \frac{\rho_1 L}{\pi (b^2 - a^2)}, R_2 = \frac{\rho_2 L}{\pi a^2}$$

 3. **Ans (D)**

$$J = 5m \times V_{cm}$$

In the COM frame, particles will be in circular motion.



$$T = \frac{m(V_{A,cm})^2}{(4l/5)} = \frac{m \left(\frac{J}{5m} \right)^2}{(4l/5)} = \frac{J^2}{20ml}$$

4. **Ans (D)**

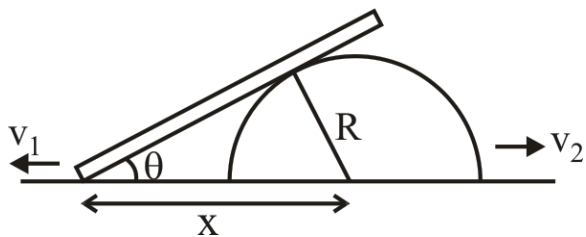
(i) At $t = 0$, capacitor is uncharged, so it behaves as zero resistance

$$i = \frac{2}{2000} = 1\text{mA}$$

(ii) Now at $t = \infty$ i.e. steady state, there is no current in capacitor branch

$$\text{so } i = \frac{2}{2000 + 2000} = 0.5\text{mA}$$

5. **Ans (A)**



$$\sin \theta = \frac{R}{x}$$

$$x = R \sec \theta$$

$$\frac{dx}{dt} = -R \sec \theta \cot \theta \frac{d\theta}{dt}$$

$$\frac{dx}{dt} = (v_1 + v_2)$$

$$\therefore (v_1 + v_2) = R \sec \theta \cot \theta \frac{d\theta}{dt}$$

$$\frac{d\theta}{dt} = \left(\frac{(v_1 + v_2) \sin^2 \theta}{R \cos \theta} \right)$$

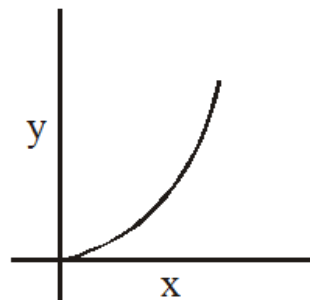
6. **Ans (C)**



Elastic energy density at

$$A = \frac{1}{2} \frac{(\text{stress})^2}{Y} = \text{constant}$$

$$\frac{1}{2} \frac{\left(\frac{M}{L} \frac{x}{A} g \right)^2}{Y} = \text{constant}$$



$$Y \propto x^{2+}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (A,C)**

$$(V_L - V_C) = 50$$

$$V_R^2 + (V_L - V_C)^2 = 130^2 \Rightarrow V_R^2 = 130^2 - 50^2$$

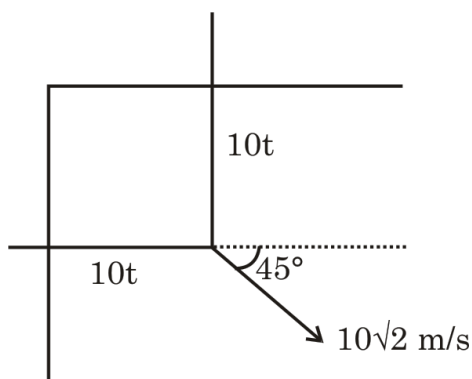
$$\therefore \text{power factor} = \frac{V_R}{V} = \frac{120}{130} = \frac{12}{13}$$

$$V_R^2 + V_L^2 = 150^2$$

$$\Rightarrow V_L = \sqrt{150^2 - 120^2} = 90 \text{ V}$$

$$V_L - V_C = 50 \Rightarrow V_C = 90 - 50 = 40 \text{ V}$$

8. Ans (A,C,D)



At time t ,

$$\phi = BA$$

$$\phi = (0.1)(10t)^2$$

$$\phi = 10t^2$$

$$\text{emf} = \frac{d\phi}{dt} = 20t$$

$$i = \frac{20t}{R} = \frac{20t}{0.01 \times (40t)} = 50A$$

9. Ans (A,B)

$$\text{At } t < 0; I_L = \frac{6}{6} = 1A$$

$$\text{At } t \gg 0; I_L = \frac{12}{3} = 4A$$

$$\therefore |f| = L|i_f - i_i| = 500 \times 10^{-3} \times 3 = 1.5 \text{ Wb}$$

$\therefore$ (A) & (B)

10. Ans (A,B,D)

$$\text{Least count} = \frac{1\text{MSD}}{100} = 0.01\text{mm}$$

$$\text{Zero error} = -1\text{mm} + 98 \text{ LC}$$

$$= -1\text{mm} + 0.98 \text{ mm}$$

$$= -0.02 \text{ mm}$$

$$\text{Diameter} = \text{Reading} - \text{Zero error}$$

$$= (2.23 \text{ mm}) - (-0.02 \text{ mm})$$

$$= 2.25 \text{ mm}$$

11. Ans (A,C,D)

$$\text{Percentage error in } h = \frac{dh}{h} \times 100 = \frac{0.3}{46.2} \times 100 = 0.6\%$$

$$\text{Percentage error in } t = \frac{0.1}{1.6} \times 100 = 6.25\%$$

$$\text{Percentage error in } g = (\% \text{ error in } h) + 2 \times (\% \text{ error in } t)$$

$$\frac{dg}{g} = 13.1\%$$

$$g = 36 \pm 5 \text{ ft/s}^2$$

12. Ans (B,C,D)

After closing switch let charge q remains on A

$$V_A = V_C$$

$$\frac{Kq}{R} + \frac{K2\theta}{2R} + \frac{K(4\theta - q)}{3R} = \frac{K6Q}{3R}$$

$$q = -\frac{\theta}{2}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (2.28)

$$p = p_0 + h\rho g + \frac{25}{r}$$

$$= 10^5 + 5 \times 10^4 + 2 \times \frac{0.07}{3.5 \times 10^{-6}} = 10^5 + 9 \times 10^4$$

$$p = \frac{\rho RT}{M}$$

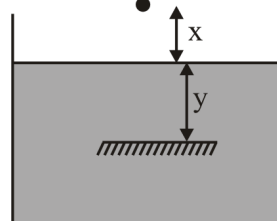
$$\frac{p}{\rho} = \frac{p'}{\rho'}$$

$$\rho' = \frac{1.9 \times 10^5 \times 1.3}{10^5} = 2.28 \text{ kg/m}^3$$

2. Ans (3.00)

Apparent distance of mirror from O

Object



$$= x + \frac{y}{\mu}$$

Distance of final image from O

$$= 2 \left(x + \frac{y}{\mu} \right)$$

velocity of image

$$= 2 \left(\frac{dx}{dt} + \frac{1}{\mu} \frac{dy}{dt} \right) = \frac{2}{\mu} \times 8 = \frac{2}{4/3} \times 8 = 12 \text{ cm/s}$$

4. **Ans (34.58)**

The number of nuclei present is given by $N = 0.05N_0$

$$N = N_0 e^{-\lambda t} \rightarrow t = -\frac{\ln N/N_0}{\lambda}$$

$$= -\frac{T_{1/2} \ln N/N_0}{0.693} = -\frac{(8.0d) \ln 0.05}{0.693} = 34.58d = \boxed{35d}$$

5. **Ans (1.98)**

$$\gamma \rightarrow -1e^{\circ} + +1e^{\circ}$$

from energy mass equivalence

$$E = m_e C^2 + M_e C^2$$

$$= 2m_e C^2 = \frac{2 \times 9.1 \times 10^{-31} \times (3 \times 10^8)^2}{1.6 \times 10^{-19} \times 10^6}$$

$$= 1.02 \text{ MeV}$$

$$\therefore \text{K.E.} = 3 - 1.02 = 1.98 \text{ MeV}$$

6. **Ans (19.00)**

$$n \frac{hc}{\lambda} = 2 \times 10^{-3} \times \frac{1}{1.6 \times 10^{-19}}$$

$$n \frac{1240}{620} = 2 \times 10^{-3} \times \frac{10^{-19}}{1.6} \Rightarrow n = \frac{1}{1.6} \times 10^{+16}$$

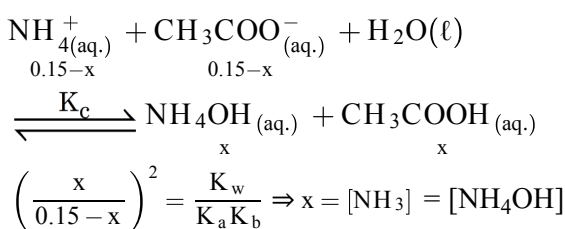
$$\text{number of e/sec} = \frac{1}{1.6}$$

$$i = 10^{-19} \text{ A}$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. **Ans (A)**



4. **Ans (B)**

$$R = Ae^{-E_a/RT}$$

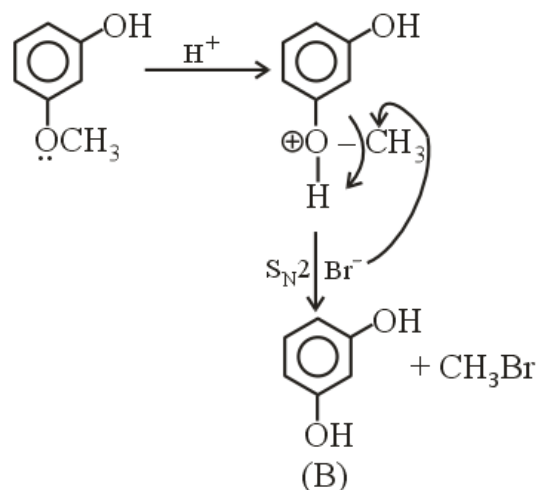
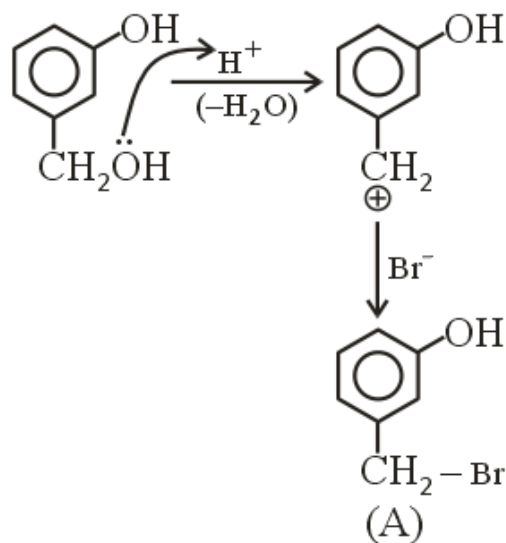
$$= 1.25 \times 10^8 e^{-\frac{16 \times 1000}{2 \times 400}} \text{ unit of A is sec}^{-1}$$

$\Rightarrow$ first order reaction

$$= \frac{1.25 \times 10^8}{e^{20}} = \frac{1.25 \times 10^8}{5 \times 10^8} = \frac{1}{4}$$

$$\text{Average life} = \frac{1}{K} = \frac{1}{1/4} = 4 \text{ sec.}$$

5. **Ans (A)**



6. **Ans (D)**

Gly-Lys is a basic peptide, it has a net positive charge at pH 6, and it therefore migrates to the cathode, the negatively charged electrode.

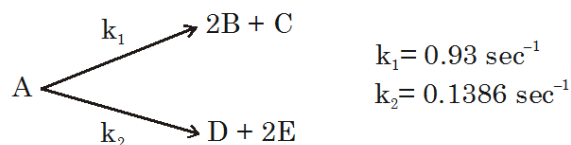
Because Gly-Asp is an acidic peptide, it has a net negative charge at pH 6, and it migrates to the anode, the positively charged electrode.

Gly-Ala is a neutral peptide, it has net zero charge at pH 6, and therefore it does not migrate

PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (A,B,C,D)



$$(t_{1/2})_1 = \frac{0.693}{0.0693} \text{ sec} = 10 \text{ sec.}$$

$$(t_{1/2})_2 = \frac{0.693}{0.1386} \text{ sec} = 5 \text{ sec.}$$

$$(t_{1/2})_{\text{overall}} = \frac{(t_{1/2})_1(t_{1/2})_2}{(t_{1/2})_1 + (t_{1/2})_2} = \frac{50}{15} \left(\frac{10}{3} \right) \text{ sec}$$

$$\frac{[B]}{[D]} = \frac{2k_1}{k_2} = 1$$

$$\begin{array}{lcl}
 A \longrightarrow 2B + C & \left. \begin{array}{l} x + y = 1 \\ \frac{2x}{y} = 1 \Rightarrow x = \frac{y}{2} \end{array} \right\} & \\
 2-x-y & 2x & x \\
 A \longrightarrow D + 2E & & \\
 2-y-x & y & 2y
 \end{array}
 \Rightarrow x = \frac{2}{3}; x = \frac{1}{3}$$

$$\Rightarrow \text{Angle of notation after } \left(\frac{10}{3} \right) \text{ sec.}$$

$$\begin{aligned}
 &= \left(\frac{2}{3} \right) \cdot (\theta_B) + (1) (\theta_A) + \left(\frac{2}{3} \right) (\theta_B) \\
 &= 60^\circ + \frac{2}{3} \times 30^\circ - \frac{2}{3} \times (-90^\circ) = 20^\circ
 \end{aligned}$$

After % decomposition

$$x + y \cdot \frac{3}{4} \times 2 = \left(\frac{3}{2} \right)$$

$$\frac{2x}{y} = 1 \Rightarrow x = \left(\frac{y}{2} \right)$$

$$\left(\frac{3y}{2} \right) = \frac{3}{2} \Rightarrow y = 1; x = \frac{1}{2}$$

Angle of rotation after first conversion

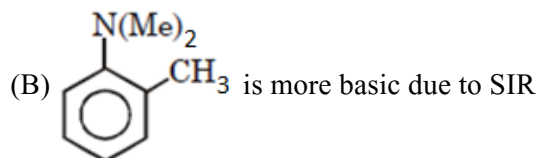
$$\begin{aligned}
 &\left(2 - \frac{3}{2} \right) \theta_A + (1) \theta_B + (1) \theta_D \\
 &= \frac{\theta_A}{2} + \theta_B + \theta_D = \frac{60^\circ}{2} + 30^\circ - 90^\circ = -90^\circ
 \end{aligned}$$

$\Rightarrow$ levorotary.

% C in the product

$$\begin{aligned}
 &= \frac{[C]}{[C] + [D] + [B] + [E]} \times 100 \\
 &= \frac{x}{x + 2x + y + 2y} \times 100 = 11.11 \%
 \end{aligned}$$

12. Ans (A,C,D)

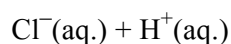
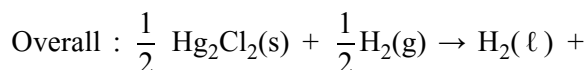
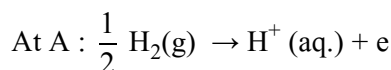
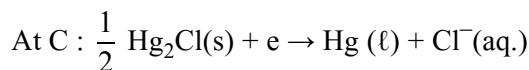


(Steric inhibition of resonance) effect

PART-2 : CHEMISTRY

SECTION-II

3. Ans (6.00)



$$E_{\text{cell}}^0 = E_{\text{Cl}^-/\text{Hg}_2\text{Cl}_2/\text{Hg}}^0 - E_{\text{H}^+/\text{H}_2}^0 = 0.28 - 0 = 0.28 \text{ V}$$

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.06}{1} \log \frac{[\text{Cl}^-]_C [\text{H}^+]_A}{P_{\text{H}_2(\text{g})}^{1/2}}$$

$$\text{or, } 0.64 = 0.28 - 0.06 \log \frac{1 \times [\text{H}^+]_A}{1}$$

$$\Rightarrow 0.36 = 0.06 \text{ pH}$$

$$\Rightarrow \text{pH} = \frac{36}{6} = 6$$

4. **Ans (990.00)**

$$V_1 = 100 \text{ ml}, V_2 = (100 - 1) = 99 \text{ ml}$$

$$W = -P_{\text{ext}}(V_2 - V_1)$$

$$= -100(99 - 100)$$

$$= 100 \text{ bar ml} = 10 \text{ J} (\because 1 \text{ bar ml} = 0.1 \text{ J})$$

$$\text{or, } W = 10 \text{ J}$$

$$\Delta U = W = 10 \text{ J} = 100 \text{ bar ml}$$

$$\Delta H = \Delta U + (P_2 V_2 - P_1 V_1)$$

$$= 100 + (100 \times 99 - 1 \times 100) = 100 + 9800 = 9900$$

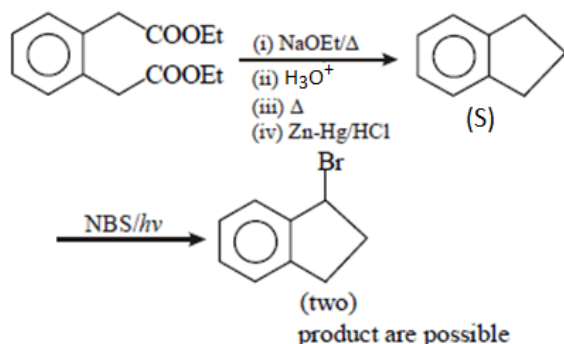
$$\text{bar ml}$$

$$= 990 \text{ J}$$

5. **Ans (7.00)**

$$3, 4, 6, 7, 8, 9, 10$$

6. **Ans (2.00)**



PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (D)**

$$T_1 = 1, 8, 15, 22$$

$$T_2 = 2, 9, 16, 23$$

$$T_3 = 3, 10, 17, 24$$

$$T_4 = 4, 11, 18, 25$$

$$T_5 = 5, 12, 19$$

$$T_6 = 6, 13, 20$$

$$T_7 = 7, 14, 21$$

$$\text{Favourable ways} = 4 \cdot {}^4C_2 (T_1, T_2, T_3, T_4)$$

$$+ 3 \cdot {}^3C_2 (T_5, T_6, T_7)$$

$$+ 2 \cdot {}^4C_1 {}^3C_1 (T_1, T_6, T_2, T_5)$$

$$+ {}^4C_1 {}^4C_1 (T_3, T_4)$$

$$= 73$$

$$\text{Probability} = \frac{73}{{}^{25}C_2} = \frac{73}{300}$$

2. **Ans (C)**

$$f(0) = 2 \text{ \& } f(x)$$

$$= (e^x + e^{-x})\cos x - 2x - \left[x \int_0^x f'(t) dt - \int_0^x t f'(t) dt \right]$$

$$\Rightarrow f(x) = (e^x + e^{-x})\cos x - \int_0^x f(t) dt$$

$$\text{or } f'(x) + f(x) = (e^x - e^{-x})\cos x - (e^x + e^{-x})\sin x$$

$$\text{Also } f'(0) + f''(0) = 0$$

$$\text{Integration factor of DE} \Rightarrow e^x$$

$$\Rightarrow ye^x = \int e^{2x} (\cos x - \sin x) dx - \int (\cos x + \sin x) dx$$

$$\Rightarrow y = e^x \left(\frac{3 \cos x}{5} - \frac{\sin x}{5} \right) - (\sin x - \cos x) e^{-x} + \frac{2}{5} e^{-x}$$

3. **Ans (B)**

$$h(x) = \phi(x+a) - \phi(x)$$

$$h(0) = r(r > 0), h(a) = -r, h(2a) = r$$

$$h(3a) = -r, h(4a) = r, h(5a) = -r, h(6a) = r$$

$$h(x) = 0 \text{ for atleast 6 roots in } x \in (0, 6a)$$

$$h'(x) = 0 \text{ has atleast 5 roots in } x \in (0, 6a)$$

$$g(x) = h(x) \cdot h'(x)$$

$$g(x) = 0 \text{ has atleast 11 roots in } x \in (0, 6a)$$

$$g'(x) = 0 \text{ has atleast 10 roots in } x \in (0, 6a)$$

4. Ans (A)

$$\int \frac{(x^2 - 1) dx}{\frac{(x^2+1)}{x^2} \cdot x^3 \sqrt{(x + \frac{1}{x} - 1)(x + \frac{1}{x} + 1)}}$$

$$\int \frac{(1 - \frac{1}{x^2}) dx}{(x + \frac{1}{x}) \sqrt{(x + \frac{1}{x})^2 - 1}},$$

put $x + \frac{1}{x} = t$

$$\int \frac{dt}{t\sqrt{t^2 - 1}} = \sec^{-1}t + c$$

5. Ans (A)

Ellipse $16x^2 + 25y^2 = 400 \Rightarrow \frac{x^2}{25} + \frac{y^2}{16} = 1$

Let the point on ellipse be P $(5\cos\theta_1, 4\sin\theta_1)$

and Q $(5\cos\theta_2, 4\sin\theta_2)$

Circle described on PQ as diameter touches

x-axis $(3, 0)$

$$5 \left(\frac{\cos\theta_1 + \cos\theta_2}{2} \right) = 3 \text{ and } 4 \left(\frac{\sin\theta_1 + \sin\theta_2}{2} \right) = r$$

$$\Rightarrow \frac{4}{5} \tan\left(\frac{\theta_1 + \theta_2}{2}\right) = \frac{r}{3}$$

$$\Rightarrow \tan\left(\frac{\theta_1 + \theta_2}{2}\right) = \frac{5r}{12}$$

$\Rightarrow$ Slope of PQ

$$= -\frac{4}{5} \cot\left(\frac{\theta_1 + \theta_2}{2}\right) = -\frac{48}{25r} = -1$$

6. Ans (A)

$$\text{L.H.S.} = \frac{2 - m(\sin 2\alpha + \sin 2\beta)}{1 - m(\sin 2\alpha + \sin 2\beta) + m^2 \sin 2\alpha \sin 2\beta}$$

$$= \frac{2 - 2m \sin(\alpha + \beta) \cos(\alpha - \beta)}{1 - 2m \sin(\alpha + \beta) \cos(\alpha - \beta) + \frac{m^2}{2} (\cos(2\alpha - 2\beta) - \cos(2\alpha + 2\beta))}$$

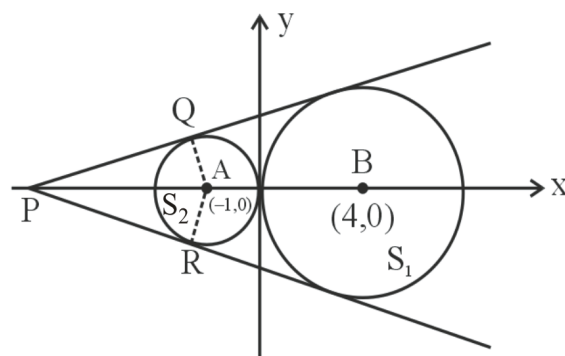
$$= \frac{2(1 - m^2 \sin^2(\alpha + \beta))}{1 - 2m^2 \sin^2(\alpha + \beta) + m^4 \sin^2(\alpha + \beta) + m^2 \sin^2(\alpha + \beta) - m^2}$$

$$= \frac{2(1 - m^2 \sin^2(\alpha + \beta))}{(1 - m^2)(1 - m^2 \sin^2(\alpha + \beta))} = \frac{2}{1 - m^2}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A,C)



$$r_2 = 1, r_1 = 4$$

$$\frac{PA}{PB} = \frac{1}{4} \therefore P \left(\frac{1 \times 4 - 4 \times -1}{1 - 4}, 0 \right)$$

$$\therefore P \left(-\frac{8}{3}, 0 \right)$$

Circum circle of ΔPQR

$$\left(x + \frac{8}{3}\right)(x + 1) + y^2 = 0$$

$$x^2 + y^2 + \frac{11}{3}x + \frac{8}{3} = 0$$

8. Ans (A,B,C,D)

Given, $|2z + 3i| = |z|^2$

$\therefore |2z + 3i| \leq 2|z| + 3$

$\Rightarrow |z|^2 \leq 2|z| + 3$

$\Rightarrow 0 \leq |z| \leq 3 \quad \dots(1)$

Again, $|2z + 3i| \geq |2|z| - 3|$

$\Rightarrow |z|^2 \geq |2|z| - 3|$

$\therefore |z| \geq 1 \quad \dots(2)$

So, (1) $\cap$ (2) gives

$1 \leq |z| \leq 3$

Also, $|z|_{\text{maximum}} \Rightarrow z = 3i$

So, $\alpha = 0, \beta = 3$

& $|z|_{\text{minimum}} \Rightarrow z = -i$

so, $x = 0, y = -1$

9. Ans (A,B)

We have $f(x)$

$$= \frac{\frac{\pi}{2} - \cos^{-1}x + \frac{\pi}{2} - \cot^{-1}x}{\cos^{-1}x + \cot^{-1}x} = \frac{\pi - \cos^{-1}x - \cot^{-1}x}{\cos^{-1}x + \cot^{-1}x}$$

$$= \frac{\pi}{\cos^{-1}x + \cot^{-1}x} - 1$$

Clearly $A = [-1, 1]$. So in $[-1, 1]$, $\cos^{-1}x \in [0, \pi]$

and $\cot^{-1}x \in \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$

$\therefore \cos^{-1}x + \cot^{-1}x = \left[\frac{\pi}{4}, \frac{7\pi}{4}\right] \forall x \in A$

Also $\cos^{-1}x + \cot^{-1}x$ is strictly decreasing on A

$$\therefore m = f(x)_{\min} = \frac{\pi}{(\cos^{-1}x + \cot^{-1}x)_{\max}} - 1$$

$$= \frac{\pi}{\cos^{-1}(-1) + \cot^{-1}(-1)} - 1 = \frac{-3}{7}$$

and $M = f(x)_{\max} = \frac{\pi}{(\cos^{-1}x + \cot^{-1}x)_{\min}} - 1$

$$= \frac{\pi}{\cos^{-1}(1) + \cot^{-1}(1)} - 1 = 3$$

Hence $\frac{M}{m} = \frac{3}{\left(-\frac{3}{7}\right)} = -7$

10. Ans (A,C)

(A) $g(x) = x^{10} - f(2x), g(0)g(1) < 0$ (A is true)

(B) $g'(x) = 2x - f(x) = 0 \quad g(0) = 0 \quad g(1) > 0$

(B is true)

(C) $g(x) = \int_0^{\frac{\pi}{2}-x} f(3t) \sin 6t \, dt - x, g(0), g(1) < 0$

(C is true)

(D) $g(0) = 0, g(1) > 0$

$g'(x) > 0$ so $g(x) \neq 0$ for $x \in (0, 1)$

11. Ans (A,B,C,D)

$$h(x) = \begin{cases} f(x) & x^2 < 1 \\ \frac{f(x)+g(x)}{2} & x^2 = 1 \\ g(x) & x^2 > 1 \end{cases}$$

$$\Rightarrow h(x) = \begin{cases} x^2 + bx + 1 & x < -1 \\ \frac{(a+2-b)}{2} & x = -1 \\ \tan \frac{(\pi(x+1))}{8} + a & -1 < x < 1 \\ \frac{1+a+2+b}{2} & x = 1 \\ x^2 + bx + 1 & x > 1 \end{cases}$$

Now, $f(-1^-) = f(-1) = f(-1^+)$

$$\Rightarrow 1 - b + 1 = \frac{a + 2 - b}{2} = a \Rightarrow a + b = 2$$

and $f(1^-) = f(1) = f(1^+)$

$$\Rightarrow 1 + a = \frac{1 + a + 2 + b}{2} = 2 + b \Rightarrow a - b = 1$$

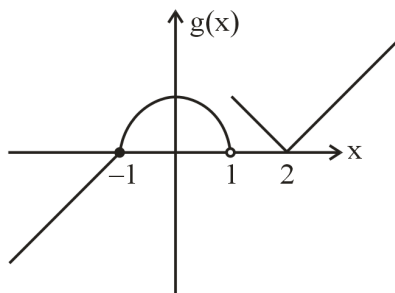
Solving, we get $a = \frac{3}{2}$ & $b = \frac{1}{2}$

$$a^2 + b^2 = \frac{9}{4} + \frac{1}{4} = \frac{5}{2}$$

$$\& g(x) \geq \frac{-D}{4a} = \frac{-(b^2 - 4)}{4} = \frac{15}{16}$$

and $f(x)$ is periodic with period $\frac{\pi}{\pi/8} = 8$

12. Ans (A,C,D)



$g(x)$ is discontinuous at $x = 1$ and

non-differentiable at $x = -1, 1, 2$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.00)

$$y = \lim_{n \rightarrow \infty} \left[\frac{(n+x)(n+2x) \dots (n+2nx)}{n^{2n}} \right]^{\frac{1}{2n}}$$

$$\left(\tan \frac{\pi}{2n} \cdot \tan \frac{2\pi}{2n} \dots \right)^{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} \left[\left(1 + \frac{x}{n}\right) \left(1 + \frac{2x}{n}\right) \dots \left(1 + \frac{2nx}{n}\right) \right]^{\frac{1}{2n}} \times$$

$$\prod_{r=1}^{n-1} \left[\tan \left(\frac{r\pi}{2n} \right) \right]^{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} \prod_{r=1}^{2n} \left(1 + \frac{rx}{n}\right)^{\frac{1}{2n}} \prod_{r=1}^{n-1} \left(\tan \frac{r\pi}{2n} \right)^{\frac{1}{n}}$$

$$\therefore \ln y =$$

$$\lim_{n \rightarrow \infty} \left\{ \frac{1}{2n} \sum_{r=1}^{2n} \ln \left(1 + \frac{rx}{n}\right) + \frac{1}{n} \sum_{r=1}^{n-1} \ln \left(\tan \frac{r\pi}{2n} \right) \right\}$$

$$= \frac{1}{2} \int_0^2 \ln(1+tx) dt + \int_0^1 \ln \left(\tan \frac{t\pi}{2} \right) dt$$

$$= \frac{1}{2x} \int_0^{2x} \ln(1+t) dt + \frac{2}{\pi} \int_0^{\pi/2} \ln \tan t dt$$

$$= \frac{1}{2x} \left[(1+t) \ln(1+t) - (1+t) \right]_0^{2x} +$$

$$\frac{2}{\pi} \int_0^{\pi/2} (\ln \sin t - \ln \cos t) dt$$

$$= \frac{1}{2x} \left[(1+2x) \ln(1+2x) - 2x \right] + \frac{2}{\pi} (0)$$

$$\Rightarrow \lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{(1+2x) \ln(1+2x)}{2x} - 1 = 0$$

$$\Rightarrow \lim_{x \rightarrow 0} y = \lim_{x \rightarrow 0} e^{\ln y} = e^0 = 1$$

2. Ans (5.00)

$$a_1 = 1, a_2 = 3, a_3 = 3, a_4 = 3, a_5 = 5, \dots a_9 = 5,$$

$$a_{10} = 7, \dots a_{16} = 7, a_{17} = 9, \dots$$

$$\Rightarrow a_{k^2} = k^{\text{th}} \text{ odd number} = 2k - 1$$

$$\text{and } a_{k^2+1} = (k+1)^{\text{th}} \text{ odd number} = 2k + 1$$

$$\Rightarrow a_{k^2+1} - a_{k^2} = 2$$

$$\Rightarrow b \left[\sqrt{k^2 + 1 + c} \right] - \left[\sqrt{k^2 + c} \right] = 2$$

$$\Rightarrow \left[\sqrt{k^2 + 1 + c} \right] - \left[\sqrt{k^2 + c} \right] = \frac{2}{b}$$

$$\Rightarrow b = 2 \text{ and } \Rightarrow \left[\sqrt{k^2 + 1 + c} \right] = 1 + \left[\sqrt{k^2 + c} \right]$$

$$\Rightarrow k^2 + 1 + c \text{ is perfect square}$$

$$\Rightarrow c = -1$$

$$\Rightarrow a_1 = 1 \Rightarrow d = 1$$

3. Ans (7.75)

Plane containing line will be given as

$$(4x - 2y - 10) + m.(5y - 4z + 3) = 0$$

$$\Rightarrow 4x + (5m - 2)y - 4mz + 3m - 10 = 0$$

The above plane is passing through (4, 3, 7)

$$\text{Hence } 16 + (5m - 2)3 - 4m.7 + 3m - 10 = 0$$

$$\Rightarrow m = 0$$

$$\text{Plane is } 4x - 2y - 10 = 0$$

If this plane is rotated an angle $\frac{\pi}{2}$ the plane

$$\text{will be } 4x + (5m - 2)y - 4mz - (3m - 10) = 0$$

$$\text{Also } 16 + (5m - 2)(-2) = 0$$

$$\Rightarrow m = 2; \text{ i.e. plane is } 4x + 8y - 8z = 4$$

$$\Rightarrow x = 2y - 2z = 1 \Rightarrow a = 1, b = 2, c = -2$$

$$\Rightarrow 2a + 3b + \frac{c}{8} = 2 + 6 - \frac{1}{4} = 7.75$$

4. Ans (20.00)

$$\text{Let the matrix be } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$\text{then } A^2 - 2A + I = 0.$$

Sum of roots is 2 & product of roots is 1, which represent trace & determinant of matrix respectively.

Then pair of (a, d) can be (5, -3), (-3, 5), (4, -2),

(-2, 4), (3, -1), (-1, 3), (2, 0) & (0, 2)

(Because sum of a + d = 2)

Now Case-I (a, d) is (5, -3) or (-3, 5)

$$\text{Here } |A| = ad - bc = 1 \Rightarrow b = -16$$

$$\Rightarrow (b, c) \text{ can be } (4, -4) \text{ or } (-4, 4)$$

$\Rightarrow$ 4 possibilities of matrix A is possible

Case-II (a, d) = (4, -2) or (-2, 4)

$$\text{Here } |A| = ad - bc = 1 \Rightarrow bc = -9$$

$$\Rightarrow (b, c) = (-3, 3) \text{ or } (3, -3)$$

$\Rightarrow$ 4 possibilities of matrix A is possible

Case-III (a, d) = (3, -1) or (-1, 3)

$$\text{Here } |A| = ad - bc = 1 \Rightarrow bc = -4$$

$$\Rightarrow (b, c) = (2, -2), (-2, 2), (1, -4), (-4, 1)$$

$\Rightarrow$ 8 possibilities of matrix A is possible

Case-IV (a, d) = (2, 0) or (0, 2)

$$|A| = ad - bc = 1 \Rightarrow bc = -1$$

$$\text{Here } (b, c) = (1, -1) \text{ or } (-1, 1)$$

$\Rightarrow$ 4 possibilities of matrix A is possible

$$\text{Total possible matrices} = 4 + 4 + 8 + 4 = 20$$

5. Ans (2.00)

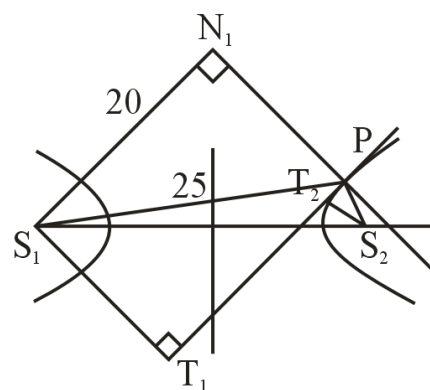
$$\lambda_1 = \frac{5!}{2!} \times {}^6C_3 - 4! \times {}^5C_3 = 1200 - 240 = 960$$

$$\lambda_2 = 3! \times {}^4C_2 \times 2 = 72$$

$$\lambda_3 = 3.4.2^3 - (1 + {}^5C_1 + ({}^2C_1 + {}^5C_2))$$

$$\lambda_3 = 96 - (6 + 2 + 10) = 78$$

6. Ans (45.00)



$$\frac{PS_1}{PS_2} = 5 \quad \dots(1)$$

$$|\text{PS}_1 - \text{PS}_2| = 20 \quad \dots(2)$$

$$\Rightarrow \text{PS}_1 = 25, \text{PS}_2 = 5$$

Now $PN_1S_1T_1$ is a rectangle

Distance of S_1 from tangent at P is

$$\sqrt{(25)^2 - (20)^2} = 15 = S_1 T_1$$

Because $\Delta PS_1 T_1 \sim \Delta PS_2 T_2$,

$$\text{so} \Rightarrow \frac{P S_1}{S_1 T_1} = \frac{P S_2}{S_2 T_2} \Rightarrow \frac{P S_1}{P S_2} = \frac{S_1 T_1}{S_2 T_2}$$

(Tangent is the internal angle bisector)

Distance of S_2 from tangent at P is $\frac{15}{5} = 3$

$$b^2 = 15 \times 3 = 45 \Rightarrow 2b = 6\sqrt{5}$$

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE ADVANCED)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,D	A,B,D	A,B	A,B	A,C	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	17.24	1.73	2.31	0.20	27.00	56.84
SECTION-III	Q.	1	2	3	4	5	6
	A.	8	5	6	5	8	7

PART-2 : CHEMISTRY

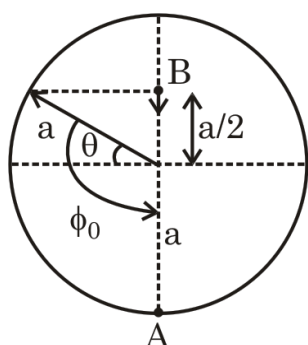
SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C,D	A,B,C	A,C,D	A,B,C	A,B,C,D	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	6.30	200.00	300.00	64.00	7.00	4.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	1	5	0	3	5	1

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,C	A,C	A,D	A,B,C,D	B,C	A,B
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.50	1.60	1.25	2.40	1.41 or 1.42	0.75
SECTION-III	Q.	1	2	3	4	5	6
	A.	1	8	9	4	5	7

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

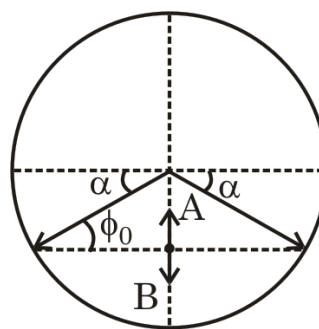
1. Ans (B,D)



$$\phi_0 = \theta + \pi/2$$

$$\theta = \pi/6$$

$$\phi_0 = \pi/6 + \pi/2 = \frac{4\pi}{6} = \frac{2\pi}{3}$$



$$\theta + \alpha = \omega \Delta t$$

$$\frac{\pi}{2} - \alpha = \omega \Delta t$$

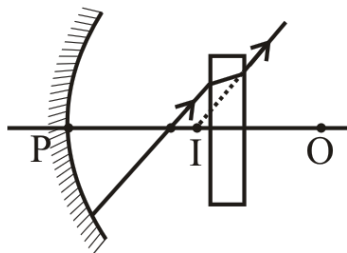
$$\theta + \frac{\pi}{2} = 2\omega \Delta t$$

$$\Delta t = \frac{\theta + \pi/2}{2\omega} = \frac{2\pi/3}{2\omega} = \frac{\pi}{3\omega}$$

2. **Ans (A,B,D)**

$$\text{Shift} = 4 \left(1 - \frac{1}{2} \right) = 2\text{cm}$$

Shift will always takes place in direction of incident ray.



So, for mirror

$$u = -150\text{ cm}$$

$$v = \frac{uf}{u-f} = -75\text{cm}$$

$$\text{final images} = 75 + 2 = 77\text{cm}$$

image will be virtual

$$PI = 77\text{ cm}$$

3. **Ans (A,B)**

$$p = q2R$$

$$\tau = pE\sin\theta$$

$$\tau = q2RE\sin\theta = I\alpha$$

$$q2RE\sin\theta = \frac{MR^2}{2}\alpha$$

For small angle

$$\alpha = \frac{4Eq}{MR}\theta$$

5. **Ans (A,C)**

$$i_1 = i_2$$

$$neA_0v_1 = \frac{neA_0v_2}{4}$$

$$\frac{v_1}{v_2} = \frac{1}{4}$$

As $E \propto v_d$

$$\frac{E_1}{E_2} = \frac{v_1}{v_2} = \frac{1}{4}$$

6. **Ans (A,B,C,D)**

If initial capacitance is C, capacitance after inserting dielectric = KC

Charge: As potential difference is same

$$Q_{in} = CV$$

$$Q_{fin} = KCV = KQ_{in}$$

As V and separation are same, E is same.

PART-1 : PHYSICS

SECTION-II

1. **Ans (17.24)**

Velocity of wave

$$\text{For string } 3\mu_0 \Rightarrow v = \sqrt{\frac{F_0}{3\mu_0}}$$

$$\text{For } \frac{\mu_0}{2} \text{ string} \Rightarrow v' = \sqrt{\frac{2F_0}{\mu_0}}$$

$$\text{Time } t = \frac{\frac{L}{2}}{\sqrt{\frac{F_0}{3\mu_0}}} + \frac{\frac{L}{2}}{\sqrt{\frac{2F_0}{\mu_0}}}$$

$$= \frac{L}{2} \sqrt{\frac{\mu_0}{3F_0}} \left(\sqrt{3} + \frac{1}{\sqrt{2}} \right)$$

On putting all the values, time $t = 17.24\text{ ms}$

2. **Ans (1.73)**

$$P_0 + \frac{\rho g H}{3} + \frac{2\rho g H}{3} + 3\rho g \left(\frac{H}{3} - h \right) = P_0 + \frac{1}{2}3\rho V^2$$

$$\Rightarrow 2gH - 3gh \dots\dots = \frac{1}{2}3V^2$$

$$\Rightarrow \frac{2}{3}g(2H - 3h) = V^2$$

$$\Rightarrow \frac{20}{3}(6 - 1.5) = V^2 = 30$$

$$x = V \sqrt{\frac{2h}{g}} = \sqrt{30} \sqrt{\frac{2 \times 0.5}{10}}$$

$$\sqrt{3} = 1.73$$

3. **Ans (2.31)**

Work energy theorem

$$W = (KE)_f - (KE)_i$$

$$\text{As } (KE)_i = 0$$

$$(\text{Bid } \sin 90^\circ) \ell = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$\text{for cylinder, } I = \frac{MR^2}{2} \text{ and for no slipping } v = \omega R$$

$$\text{Bild} = \frac{1}{2}mv^2 + \frac{1}{2} \left(\frac{MR^2}{2} \frac{v^2}{R^2} \right)$$

$$\text{Bild} = \frac{1}{2}mv^2 \left[1 + \frac{1}{2} \right] = \frac{3}{2} \times \frac{1}{2}mv^2$$

Putting values

$$4 = \frac{3}{4}v^2$$

$$v = \frac{4}{\sqrt{3}} = 2.31\text{ m/s}$$

4. Ans (0.20)

$$\lambda = 6000 \text{ \AA}, \mu_g = 1.5$$

maximum intensity without film

$$I_{\max} = I_0 + I_0 + 2\sqrt{I_0 I_0} \cos \theta^\circ = 4I_0$$

$$75\% \text{ of maximum means } 0.75 \times 4I_0 = 3I_0$$

$$3I_0 = I_0 + I_0 + 2\sqrt{I_0 I_0} \cos \delta,$$

δ = phase difference

$$\cos \delta = \frac{1}{2}$$

$$\delta = (2n + 1) \frac{\pi}{3} \quad \dots(i)$$

$$\text{As } \delta = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{\lambda} (\mu - 1)t \quad \dots(ii)$$

Equating values of δ from both equations,

$$t = \frac{\lambda(2n + 1)}{6(\mu - 1)} = \frac{6000 \times 10^{-10}}{6(1.5 - 1)} (2n + 1)$$

For min t , n should be 0.

$$t = \frac{10^{-7}}{0.5} \times 1 = 0.2 \mu\text{m}$$

$$t = 0.2 \mu\text{m}$$

5. Ans (27.00)

Power incident = power lost

$$1944 \times \pi R^2 = \sigma \times 4\pi R^2 (T^4)$$

$$1944 \times \frac{1}{4} = \sigma T^4$$

$$\frac{1944}{4 \times 6 \times 10^{-8}} = 81 \times 10^8 = T^4$$

$$T = 300 \text{ K or } 27^\circ\text{C}$$

6. Ans (56.84)

$$\text{Maximum energy of incident light} = \frac{hc}{\lambda_{\min}}$$

$$E_{\max} = \frac{hc}{400\text{nm}} = \frac{1240 \text{ eV} \cdot \text{nm}}{400 \text{ nm}} = 3.1 \text{ eV}$$

$$(KE)_{\max} \text{ of emitted electrons} = 3.1 - 2.39 = 0.71 \text{ eV}$$

$$\text{For no current, } 2r = 10 \text{ cm} \Rightarrow r = 5 \text{ cm}$$

$$r = \frac{mv}{qB} \Rightarrow 5 \times 10^2$$

$$r = \frac{\sqrt{2mKE}}{qB} = \frac{\sqrt{2 \times 9.1 \times 10^{-31} \times 0.71 \times 1.6 \times 10^{-19}}}{1.6 \times 10^{-19} \times B}$$

$$\frac{5}{100} = \frac{\sqrt{2 \times 9.1 \times 10^{-31} \times 0.71 \times 1.6 \times 10^{-19}}}{1.6 \times 10^{-19} \times B}$$

$$B = 56.84$$

PART-1 : PHYSICS

SECTION-III

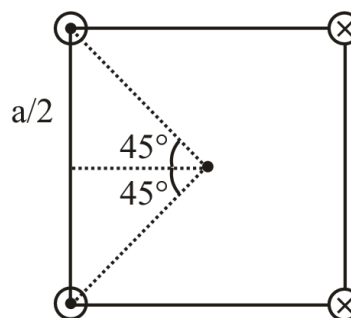
1. Ans (8)

Horizontal components of forces of all 4 wires = 0,

vertical components will be added.

$$\text{Hence, } B_{\text{net}} = 4 \times B \cos 0^\circ$$

Side view



$$B_{\text{net}} = 4B \cos 45^\circ = 4 \times \frac{\mu_0 i}{2\pi r} \times \frac{1}{\sqrt{2}}$$

$$B_{\text{net}} = \frac{4 \times 10^{-7} \times 2 \times 2}{\frac{a}{2} \sqrt{2}} \times \frac{1}{\sqrt{2}}$$

$$B_{\text{net}} = \frac{16 \times 10^{-7}}{10 \times 10^{-2}} = 16 \mu\text{T}$$

2. Ans (5)

Using parallel and series combination

$$L_{eq} = 5$$

3. Ans (6)

$$\text{Speed of Amit : } \frac{500}{V_A} = 4 \text{ min}$$

$$\Rightarrow v_A = \frac{500}{4} = 125 \text{ m/min}$$

Speed of V.B

$$v_B = \frac{700}{4} = 175 \text{ m/min}$$

Next time when Amit will reach at A

$$\Rightarrow \frac{1200}{125}, \frac{2400}{125}, \frac{3600}{125}$$

Time when Sumit B reaches

$$\frac{1200}{175}, \frac{2400}{175}, \frac{3600}{175}$$

Let Amit completed n_1 round & Sumit

completed n_2

to meet again i.e

$$\left(\frac{1200}{125} \right) n_1 = \left(\frac{1200}{125} \right) n_2$$

$$\frac{n_1}{n_2} = \frac{5}{7}$$

Next time they will meet when $n_1 = 5$ i.e.

$$t = \frac{1200}{125} \times 5$$

$$\frac{t}{8} = \frac{1200 \times 5}{125} \times \frac{1}{8} = 6 \text{ min}$$

4. Ans (5)

$$\tan \phi = \frac{\omega L_1}{R_1} = \frac{3}{4}$$

$$\omega L_1 = \frac{3}{4} R_1 \quad \dots(i)$$

$$\text{and current } I_L = 1 = \frac{100}{\sqrt{R_1^2 + (\omega L_1)^2}} = \frac{100}{\sqrt{R_1^2 + \frac{9}{16} R_1^2}}$$

$$1 = \frac{100}{\sqrt{\frac{25R_1^2}{16}}} = \frac{100 \times 4}{5R_1}$$

$$\pi R_1 = 80\Omega \text{ \& } \omega L_1 = 3/4 \times 80 = 60\Omega$$

$$\frac{\omega L_2}{R_2} = \frac{4}{3} \Rightarrow \omega L_2 = \frac{4}{3} R_2$$

$$P_2 = 5 = \frac{100}{\sqrt{R_2^2 + (\omega L_2)^2}}$$

$$5 = \frac{100}{\sqrt{R_2^2 + \left(\frac{4}{3} R_2\right)^2}} = R_2 = 12\Omega$$

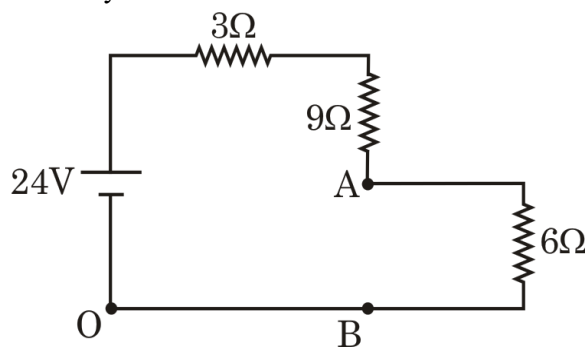
$$R_2 = 12\Omega \text{ and } \omega L_2 = 16\Omega$$

$$\tan \alpha = \frac{\omega L_1}{R_2} = \frac{60}{12} = 5$$

$$\alpha = \tan^{-1} 5$$

5. Ans (8)

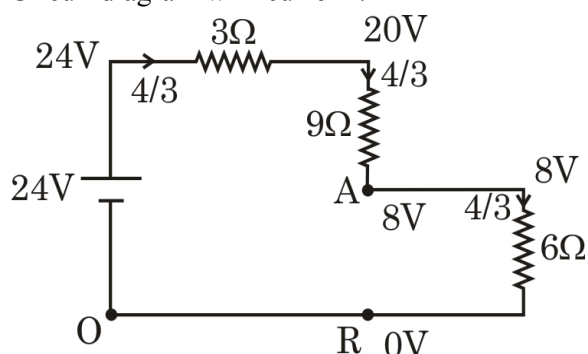
In steady circuit



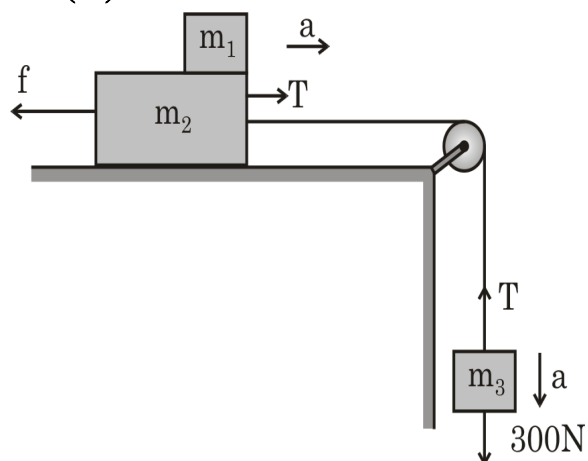
$$P = \frac{24}{18} = \frac{4}{3} \text{ A}$$

Voltage across $C_2 = V_{AB}$

Circuit diagram with current :



6. Ans (7)



For m_3

$$300 - T = m_3 a \Rightarrow 300 - T = 30a \dots (i)$$

For m_2

$$T - f = m_2 a \Rightarrow T - f = 10a \dots (ii)$$

From (i) & (ii)

$$300 - f = 40a$$

$$300 - \mu mg = 40a$$

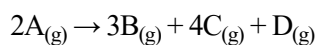
$$300 - 0.4 \times 5 \times 10 = 40a$$

$$\Rightarrow a = 7 \text{ m/s}^2$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (B,C,D)



initial moles of "A" = 10 mole

and "A" dissociates according to first order reaction.

$$\text{For "A": } 10 \text{ moles} \xrightarrow{(t_{1/2})_1} 5 \text{ moles} \xrightarrow{(t_{1/2})_2}$$

$$2.5 \text{ moles} \xrightarrow{(t_{1/2})_3} 1.25 \text{ moles} \xrightarrow{(t_{1/2})_4} 0.625 \text{ moles}$$

for first order reaction : $t_{1/2} = \text{constant}$

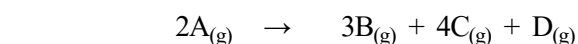
$$(t_{1/2})_1 = (t_{1/2})_2 = (t_{1/2})_3 = (t_{1/2})_4 = t_{1/2}$$

$$\Rightarrow 4.t_{1/2} = 120 \text{ min.} \Rightarrow t_{1/2} = 30 \text{ min.} = \frac{1}{2} \text{ hr.}$$

$$K_A = \frac{0.693}{1/2} = 1.386 \text{ hr}^{-1}$$

$\Rightarrow$ Rate constant of reaction

$$(K) = \frac{K_A}{2} = \frac{1.386}{2} = 0.693 \text{ hr}^{-1}$$



Initial 10 mole - - -

$$\text{at } t = 30 \text{ min. } (10 - 5) \text{ moles } \frac{3}{2} \times 5 \quad \frac{4}{2} \times 5$$

$$\frac{1}{2} \times 5$$

$$= 7.5 \text{ moles}$$

$$\text{at } t = 60 \text{ min. } (10 - 7.5) \text{ moles } \frac{3}{2} \times 7.5 \quad \frac{4}{2} \times 7.5$$

$$\frac{1}{2} \times 7.5$$

$$= 15 \text{ moles}$$

$$\text{After very long time } - \frac{3}{2} \times 10 \quad \frac{4}{2} \times 10 \quad \frac{1}{10}$$

After very long time total moles = $n_A + n_B + n_C + n_D$

$$= 0 + \frac{3}{2} \times 10 + \frac{4}{2} \times 10 + \frac{1}{10}$$

$$= 40 \text{ moles}$$

2. Ans (A,B,C)

$$10 \text{ ml } 0.3 \text{ M } \text{Na}_2\text{SO}_4 \quad \text{Na}^+ = 6 \quad \text{SO}_4 = 3$$

$$20 \text{ ml } 0.3 \text{ M } \text{Ca}(\text{NO}_3)_2 \quad \text{Ca}^{2+} = 6 \quad \text{NO}_3 = 12$$

$$\text{and } 0.3 \text{ M } \text{Sr}(\text{NO}_3)_2 \quad \text{Sr}^{2+} = 6 \quad \text{NO}_3^- = 12$$

$$K_{sp}(\text{SrSO}_4) = 10^{-7} \quad K_{sp}(\text{CaSO}_4) = 10^{-5}$$

$$\text{Total vol.} = 30 \text{ ml} \quad [\text{Ca}^{2+}] = \frac{6}{30} = 0.2 \text{ M}$$

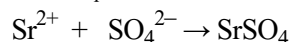
$$[\text{Sr}^{2+}] = \frac{6}{20} = 0.2 \text{ M}$$

$$\text{First ppt will be of } \text{SrSO}_4 \quad \text{SO}_4^{2-} = \frac{3}{30} = 0.1 \text{ M}$$

$$\text{Present I.P. of } \text{SrSO}_4 = [\text{Sr}^{2+}][\text{SO}_4^{2-}] = (0.2)(0.1) = 2 \times 10^{-2}$$

Now $[\text{Sr}^{2+}]$ & $[\text{SO}_4^{2-}]$ will ppt. out until

$$\text{I.P.} = K_{sp} \text{ of } \text{SrSO}_4$$



$$0.2 \text{ M} \quad 0.1 \text{ M}$$

$$(0.2-x) \quad (0.1-x) \quad x$$

$$(0.2-x)(0.1-x) = 10^{-7} \dots\dots\dots (1)$$

Maximum sulphate which can be ppt. out is 0.1 M

$$\text{Minimum value of } \text{Sr}^{2+} = 0.2 - 0.1 = 0.1 \text{ M}$$

$$\Rightarrow \text{Maximum value of } [\text{SO}_4^{2-}]$$

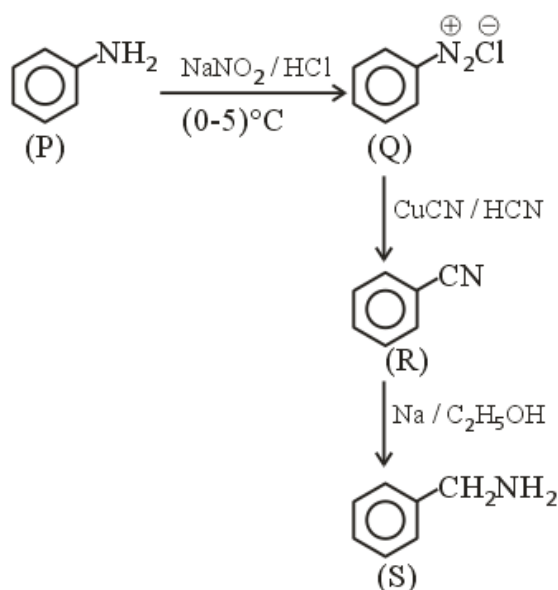
$$= \frac{K_{sp}[\text{SrSO}_4]}{[\text{Sr}^{2+}]} = \frac{10^{-7}}{0.1} = 10^{-6} \text{ M}$$

At this point $[\text{SO}_4^{2-}] = 10^{-6} \text{ M}$, $[\text{Ca}^{2+}] = 0.2 \text{ M}$
and therefore SO_4^{2-} and Ca^{2+} will not undergo ppt.

$$\text{I.P. } (\text{CaSO}_4) < K_{sp}(\text{CaSO}_4)$$

$$\Rightarrow [\text{Ca}^{2+}] = 0.2 \text{ M} \text{ is unchanged.}$$

5. Ans (A,B,C,D)



6. Ans (A,C)

(A) L-sorbose is a ketohexose.

(B) This compound is epimeric to L-sorbose at carbon-5, and thus it is not a form of L-sorbose.

In fact, it is D-fructose.

(C) This is the α -furanose form of L-sorbose (α -L-sorbofuranose)

(D) This structure is the enantiomer of the structure in option (C), and thus it is not a form of L-sorbose.

(It is the α -furanose form of D-sorbose)

PART-2 : CHEMISTRY

SECTION-II

1. Ans (6.30)

$$\Delta S = nC_{v,m} \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1} = 6.3$$

$$\frac{T_2}{T_1} = \frac{P_2 V_2}{P_1 V_1} ; (C_{v,m} = 5/2 \times R)$$

2. Ans (200.00)

$$\Delta T_f = i K_f m = K_f \cdot \frac{w_B \times 1000}{M \times w_A}$$

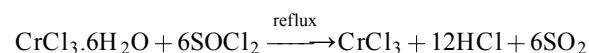
$$0.093 = 1.86 \times \frac{1}{M} \times \frac{1000}{100}$$

$$M = \frac{18.6}{0.093} = 200$$

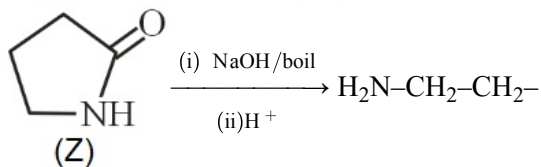
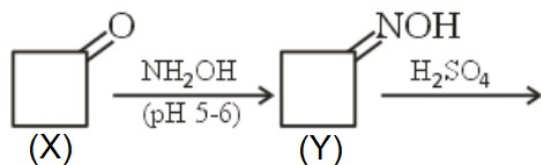
3. Ans (300.00)

($\text{NO}_2^+ \text{ NO}_3^-$ is nitronium nitrate)

4. Ans (64.00)



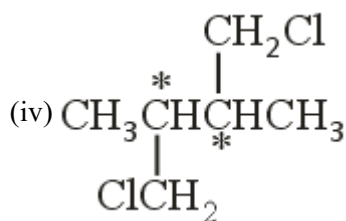
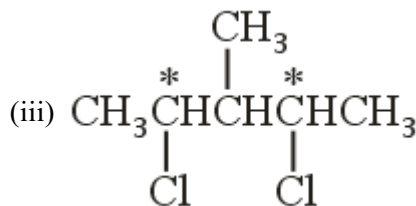
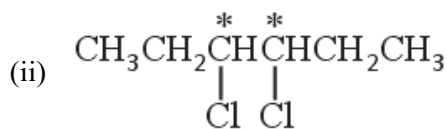
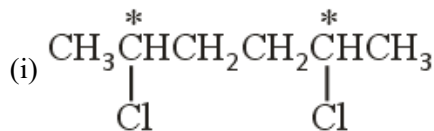
5. Ans (7.00)



CH₂-COOH

Molar mass of compound X is 70. So x = 7

6. Ans (4.00)



PART-2 : CHEMISTRY

SECTION-III

1. Ans (1)

$$E_{\text{cell}}^0 = 0 + 2.955 = 2.955$$

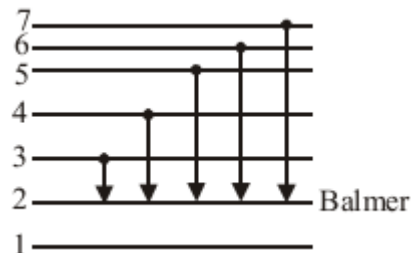
$$E^0 = \frac{0.0591}{2} \log K = 2.955$$

$$0.0591 \log K = 5.91$$

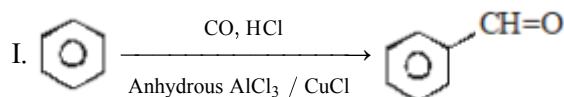
$$\log K = 100$$

$$10^{-2} \times \log K = 1$$

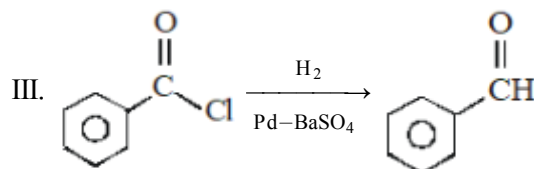
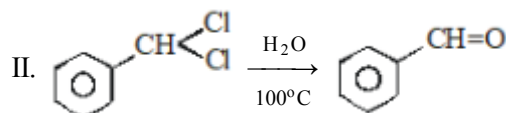
2. Ans (5)



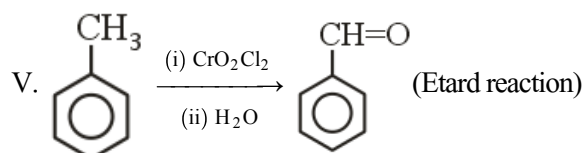
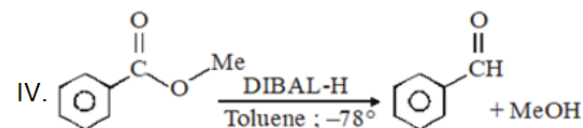
5. Ans (5)



this reaction is called Gattermann
koch synthesis



this reaction is called Rosenmund reduction



PART-3 : MATHEMATICS

SECTION-I

1. Ans (A,C)

$$1 + x > 1 + x^{\pi/2} > 1 + x^2$$

$$\int_0^1 \frac{1}{1+x^2} > \int_0^1 \frac{1}{1+x^{\pi/2}} > \int_0^1 \frac{1}{1+x}$$

$$\frac{\pi}{4} > I > \ln 2$$

2. Ans (A,C)

Let

$$P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}; P \begin{bmatrix} 2 \\ -3 \end{bmatrix} = \begin{bmatrix} -11 \\ 6 \end{bmatrix} \Rightarrow \begin{aligned} 2a - 3b &= -11 \\ 2c - 3d &= 6 \end{aligned}$$

also

$$P \begin{bmatrix} -11 \\ -6 \end{bmatrix} = \begin{bmatrix} -7 \\ -12 \end{bmatrix} \Rightarrow \begin{aligned} -11a - 6b &= -7 \\ -11c - 6d &= -12 \end{aligned}$$

$$\Rightarrow a = -1, b = 3, c = 0, d = 2$$

3. Ans (A,D)

Let's define some events : $E_1 \rightarrow$ first toss is heads.

$E_2 \rightarrow$ first two tosses are TH. & $E_3 \rightarrow$ first two tosses are TT.

$$P(A_2) = P(E_3) = \left(\frac{1}{3}\right)^2 = \frac{1}{9}. \{P(A_n) = P_n\}.$$

$$\text{For } n > 2 \rightarrow P\left(\frac{A_n}{E_1}\right) = P(A_{n-1}),$$

$$P\left(\frac{A_n}{E_2}\right) = P(A_{n-2}), P\left(\frac{A_n}{E_3}\right) = 0$$

$$\text{Let } P(A_n) = P_n$$

So, using total probability theorem :

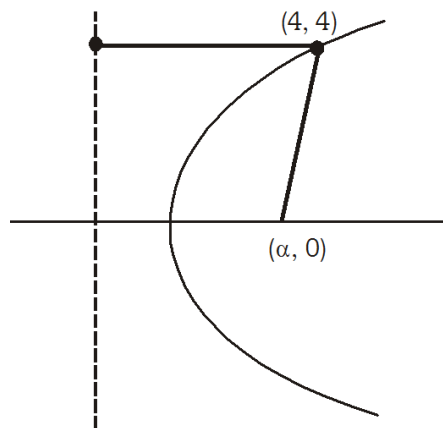
$$P_n = \frac{2}{3} \cdot P_{n-1} + \frac{2}{9} \cdot P_{n-2}; n > 2$$

$$P_1 = 0 \text{ \& } P_2 = \frac{1}{9} \Rightarrow P_3 = \frac{2}{27}, P_4 = \frac{2}{27}$$

4. Ans (A,B,C,D)

$$SP = PM$$

$$\sqrt{(4-\alpha)^2 + (4-0)^2} = \frac{|4\ell + 4m - 1|}{\sqrt{\ell^2 + m^2}}$$



$$\ell x + my = 1$$

$$\Rightarrow (\ell^2 + m^2) ((\alpha - 4)^2 + 16) = (4(\ell + m) - 1)^2$$

$$\Rightarrow (\ell^2 + m^2) (\alpha^2 + 32 - 8\alpha) = 16(\ell + m)^2 + 1 - 8(\ell + m)$$

$$\Rightarrow (\ell^2 + m^2) (\alpha^2 + 32 - 8\alpha - 16) + 8(\ell + m) - 32\ell m = 1$$

$$\Rightarrow (\ell^2 + m^2) (\alpha - 4)^2 + 8(\ell + m) - 32\ell m - 1 = 0$$

But given that

$$(\ell^2 + m^2) + 8(\ell + m) - 32\ell m - 1 = 0$$

$$\therefore \alpha = 5 \text{ or } 3$$

Take focus (3, 0)

Equation of directrix $\ell x + my = 1$ satisfies (0, 3)

i.e. mirror image of focus through tangent $y = x$

$$0 + 3m = 1$$

slope of line joining (0, 3) and (4, 4) is $\frac{1}{4}$ which is parallel to axis.

$$\therefore \text{Equation of axis } y - 0 = \frac{1}{4}(x - 3)$$

$$\Rightarrow x - 4y - 3 = 0$$

5. Ans (B,C)

$$x = (9 + 4\sqrt{5})^n = I + f \text{ where } 0 \leq f < 1$$

$$\text{Let } F = (9 - 4\sqrt{5})^n \quad 0 \leq F < 1$$

$$I + F + f = \text{even integer}$$

$$0 \leq F + f < 2 \Rightarrow F + f = 1$$

$$\Rightarrow I \rightarrow \text{odd integer}$$

$$f = \{x\} = 1 - F$$

$$\text{So } (1 - \{x\})(x) = 1$$

6. **Ans (A,B)**

Given D.Eⁿ is

$$\frac{dy}{2x dx} + \frac{y}{x^2} = \frac{\sin^{-1} x}{2x^2}$$

$$\Rightarrow \frac{dy}{dx} + \left(\frac{2}{x}\right)y = \frac{\sin^{-1} x}{x}$$

$$\text{I.F.} = e^{\int \frac{2}{x} dx} = e^{2\ln x} = x^2$$

solution is given by

$$y \times x^2 = \int \frac{\sin^{-1} x}{x} \times x^2 dx$$

$$\Rightarrow y \times x^2 = \frac{1}{2}(2x^2 - 1) \sin^{-1} x + \frac{1}{4} x \sqrt{1-x^2} + C$$

$$\text{at } x = 0, y = 0 \Rightarrow C = 0$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (1.50)**

1^∞ form

$$\Rightarrow e^{\lim_{x \rightarrow 0} \frac{4}{5x^2} (\alpha \tan \beta x^2 + 5 - 4\cos^2 x)} = e^2$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{16\sin^2 x}{5x^2} + \frac{4\alpha \sin \beta x^2}{5x^2} = 2$$

$$\Rightarrow \frac{4\alpha\beta}{5} + \frac{16}{5} = 2 \Rightarrow \alpha\beta = -\frac{3}{2}$$

2. **Ans (1.60)**

$$f(x^5 + 4x^3 + 3x) = x$$

$$\Rightarrow (5x^4 + 12x^2 + 3)f'(x^5 + 4x^3 + 3x) = 1$$

$$\text{Put } x = 1 \Rightarrow f'(8) = \frac{1}{20}$$

$$\Rightarrow 32f'(8) = 1.60$$

3. **Ans (1.25)**

Case -I : If A = 0 then number of ways = ${}^{20}C_1$

Case -II : If A = 2, then number of ways = ${}^{18}C_1$

:

Case X : If A = 18 then number of ways = 2C_1

$\therefore$ Total number of ways = ${}^{20}C_1 + {}^{18}C_1 + \dots + {}^2C_1$

Hence, K = 110

4. **Ans (2.40)**

$$x^3 - 6x^2 + kx - 80 = 0$$

α, β, γ are roots

$\alpha\beta, \beta\gamma, \gamma\alpha$ apply AM $\geq$ GM

$$\frac{\alpha\beta + \beta\gamma + \gamma\alpha}{3} \geq (\alpha^2\beta^2\gamma^2)^{1/3}$$

$$k \geq 3(\alpha\beta\gamma)^{2/3}$$

$$k \geq 3(8)^{2/3}; k \geq 12$$

5. **Ans (1.41 or 1.42)**

Substitute $x = t^2$ to get

$$I = 2 \int \frac{(t^2 - 1)}{(1 + t + t^2) \sqrt{t(1 + t^2)}} dt$$

$$= 2 \int \frac{\left(1 - \frac{1}{t^2}\right)}{\left(1 + t + \frac{1}{t}\right) \sqrt{t + \frac{1}{t}}} dt$$

$$\text{Substitute } \sqrt{t + \frac{1}{t}} = u$$

6. **Ans (0.75)**

$$C = 1 + \frac{\cos x}{3} + \frac{\cos 2x}{3^2} + \dots + \infty$$

$$S = 0 + \frac{\sin x}{3} + \frac{\sin 2x}{3^2} + \dots - \infty$$

$$C + iS = 1 + \frac{e^{ix}}{3} + \frac{e^{i2x}}{3^2} + \dots \infty$$

$$C + iS = \frac{1}{1 - \frac{e^{ix}}{3}} = \frac{3}{(3 - \cos x) - i \sin x}$$

$$C + iS = \frac{3((3 - \cos x) + i \sin x)}{(3 - \cos x)^2 + \sin^2 x}$$

$$C + iS = \frac{(9 - 3 \cos x)}{10 - 6 \cos x} + \frac{3i \sin x}{10 - 6 \cos x}$$

$$C = \frac{9 - 3 \cos x}{10 - 6 \cos x} = \frac{1}{2} \left(1 + \frac{8}{10 - 6 \cos x}\right)$$

$$C \in \left[\frac{3}{4}, \frac{3}{2}\right]$$

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (1)**

Number of terms = 3, 5, 7, 9, 11, 13, 15, 17, 19, 21

21 terms in 10th bracket.

1, 3, 6, 10, 15, 21, 28,

$$T_r = ar^2 + br + c$$

$$1 = a + b + c \quad \dots(1)$$

$$3 = 4a + 2b + c \quad \dots(2)$$

$$6 = 9a + 3b + c \quad \dots(3)$$

$$(2) - (1) \Rightarrow 2 = 3a + b$$

$$(3) - (2) \Rightarrow 3 = 5a + b$$

$$a = \frac{1}{2}, \quad b = \frac{1}{2}, \quad c = 0$$

$$T_r = \frac{r(r+1)}{2}$$

$$T_{10} = \frac{10 \times 11}{2} = 55$$

$$A = \text{Sum} = 55^1 + \dots + 55^{21}$$

$$= \frac{55(55^{21} - 1)}{54} = \frac{55^{23} - 55}{54}$$

2. **Ans (8)**

$$\Delta = 0 \text{ \& } a_0 + a_1 + a_2 \neq 0$$

$$\text{then } a_0 = a_1 = a_2$$

$$\Delta = - [(a_0^3 + a_1^3 + a_2^3) - 3a_0a_1a_2]$$

putting $x = 0$

$$a_0 = 1$$

$$\text{coefficient of } x = 4a$$

$$\text{coefficient of } x^2 = 4b + 6a^2$$

$$1 = 4a = 6a^2 + 4b$$

$$\Rightarrow a = \frac{1}{4}; \quad b = \frac{5}{32} \quad 5 \times \left(\frac{1/4}{5/32} \right) = 8$$

3. **Ans (9)**



$$\text{For A : } \left. \begin{array}{l} -2 = 1 - 3t^2 \\ 2 = t - 3t^3 \end{array} \right\} \Rightarrow t = -1$$

Equation of tangent at A :

$$y - 2 = \left(\frac{1 - 9t^2}{-6t} \right) (x + 2), \text{ where } t = -1$$

$$\Rightarrow 4x + 3y + 2 = 0$$

Pass through B $(1 - 3t_1^2, t - 3t_1^3)$

$$4(1 - 3t_1^2) + 3(t_1 - 3t_1^3) + 2 = 0$$

$$\Rightarrow (t_1 + 1)^2 (3t_1 - 2) = 0 \Rightarrow t_1 = \frac{2}{3}$$

$$\therefore B \left(-\frac{1}{3}, -\frac{2}{9} \right)$$

4. **Ans (4)**

$$f(x) = f(6 - x) \quad \dots(i)$$

$$f'(x) = -f'(6 - x) \quad \dots(ii)$$

Putting $x = 0, 2, 5$

$$f'(0) = f'(6) = f'(2) = f'(4) = f'(5) = f'(1) = 0$$

and since $f'(3) = -f'(3)$

$$\Rightarrow f'(3) = 0$$

So $f'(x) = 0$ has minimum seven (7) roots in $[0, 6]$

Now if $h(x) = f'(x)f''(x)$

$$h'(x) = (f''(x))^2 + f'(x)f'''(x)$$

$h(x) = 0$ has minimum $7 + 6 = 13$ roots so $h'(x)$

will have minimum 12 zeros is $[0, 6]$

$$\text{So } \frac{12}{3} = 4$$

5. **Ans (5)**

$$2^{2\sin^2 x - 3\sin x + 1} + 2^{3\sin x - 2\sin^2 x + 2} = 9$$

$$\Rightarrow 2^{2\sin^2 x - 3\sin x + 1} + 2^{2 - (2\sin^2 x - 3\sin x)} = 9$$

$$2^{2\sin^2 x - 3\sin x} = t$$

$$2t + \frac{4}{t} = 9 \Rightarrow t = 4 \text{ or } t = \frac{1}{2}$$

$$\Rightarrow 2^{2\sin^2 x - 3\sin x} = 2^2 \text{ or } 2^{2\sin^2 x - 3\sin x} = 2^{-1}$$

$$\Rightarrow 2\sin^2 x - 3\sin x = 2$$

$$\text{or } 2\sin^2 x - 3\sin x = -1$$

$$\Rightarrow \sin x = -\frac{1}{2}, 1, \frac{1}{2}, 2 \text{ (reject)}$$

5 solutions in $[0, 2\pi]$

6. **Ans (7)**

$$P\left(\frac{B}{A \cap C}\right) = \frac{P(A \cap B \cap C)}{P(A \cap C)}$$

$$A \cap C : (\underline{\quad} \underline{T} \underline{\quad} \underline{\quad} \underline{\quad} \underline{\quad}) \text{ or } (\underline{\quad} \underline{\quad} \underline{\quad} \underline{\quad} \underline{\quad})$$

$$\underline{T} \underline{\quad} \underline{\quad}) - (\underline{\quad} \underline{T} \underline{\quad} \underline{T} \underline{\quad} \underline{\quad})$$

$$P(A \cap C) = {}^5C_4 \cdot \left(\frac{1}{2}\right)^5 \cdot \frac{1}{2} +$$

$${}^5C_4 \cdot \left(\frac{1}{2}\right)^5 \cdot \frac{1}{2} - \left(\frac{1}{2}\right)^6 = \frac{9}{2^6}$$

$$A \cap B \cap C : (\underline{H} \underline{T} \underline{H} \underline{H} \underline{H} \underline{T}) \text{ or } (\underline{H} \underline{H} \underline{H} \underline{T} \underline{H} \underline{T})$$

$$P(A \cap B \cap C) = \frac{2}{2^6}$$

$$\text{Required probability} = \frac{\frac{2}{2^6}}{\frac{9}{2^6}} = \frac{2}{9} = \frac{m}{n}$$

$$\therefore n - m = 7$$